

by shifting repetitive or accuracy-major tasks to the platform. Currently, Amelia has been taken up by over 50 organisations globally in sectors as wide-ranging as financial services, telecommunications, leisure, travel, healthcare, IT, BPO and government. It is substantially involved in the roles of customer service, finance and banking, insurance, IT infrastructure management, human resources, service desks, or even technical guide or advisor for field-workers or engineers.

Let's take the case of an online gaming major which needed a sophisticated agent to connect with gamers for support and block phishing attempts or ever-evolving fraud techniques, for which the company needed to quickly adapt in order to maximise customer service and workflow efficiency. Amelia cannot be socially engineered—a feature the company leveraged to interact with human customers and verify accounts. Amelia is helping the gaming company to verify user identities at 100 per cent accuracy, cut fraud-screening time from five minutes down to three minutes and receive an 86.72 per cent customer satisfaction rating—higher than with human operators.

Microsoft Azure

Microsoft's machine learning platform was introduced for public preview in July 2014, and since then it has picked up good steam. The specialty of the Azure Machine Learning platform is the interactive user interface brought by the integrated Azure Machine Learning Studio, which allows users to set up their machine learning model through simple drag-and-drop actions. Inbuilt algorithm makes building supervised and unsupervised learning models easy. However, the modelling process requires active intervention of users in all development steps. Support for R and Python programming languages is a bonus for users who plan to create or customise their own ML models.

Additionally, Azure ML supports a wide range of statistical functions and methods which are core to Data Science. Supported ML models include regression, binary and multi-class classification, clustering, recommendations and anomaly detection.

All user categories ranging from SMEs up to large industries in different verticals including retail, insurance, finance and banking, healthcare, oil and gas or even manufacturing can benefit from this AI platform. Smarter and swift predictive analytics, project outcome management, re-usable and scalable models, efficient data pipelines, risk and fraud management, and patient recovery and treatment prediction are some of the biggest benefits of Azure Machine Learning.

The Microsoft Azure website showcases a story where Mendeley, a free research content reference and academic network platform for researchers, students and academic readers, implemented Azure Machine Learning to manage its users better, provide more accurate recommendations, and improve interface response and overall user experience. Feeding historical data like user activity trends, views, libraries, searches and data provided by the marketing team, helped the company to build ML models, which eventually improved the company's recall by 30 per cent.

IBM Watson

IBM Watson is an interactive cognitive platform whose analytical basis is the humongous amount of disparate big data structures that the system can engulf. It computes this data at very high speed and analyses to give the results in minimal time.

Talking about its cognitive and interactive capability, Dr Prashant Pradhan, executive director, Watson & Cloud, and chief developer advocate, IBM India/South Asia, says, "Watson understands the nuances of human language, so it is able to

bring back relevant answers in context of the question. Watson also gets smarter, learning from each interaction with its users, and each piece of data it ingests."

Driven on the IBM Bluemix cloud platform, IBM Watson comes with 50 AI APIs including IBM dialogue, retrieve and rank, machine learning, speech-to-text, text-to-speech and, more interestingly, concept and visual Insight that enable an application to expand and relate concepts, drawing on the meaning of a word rather than simple text matching, and allow developers to build apps that reveal insights from social media images and video. As an analytics platform, Watson provides a visual representation of the business drivers and insights into any focus areas. While the major involvement of the user is in loading the correct data, rest of the background computations are mostly automated within the system.

Watson has seen diverse uses in sectors like healthcare (to suggest treatments for cancer), financial services (to recommend personalised portfolios using the latest information), customer engagement, law, retail, fashion and education.

In an interesting application from India, Sriram Raghavan, director, India Research Labs, IBM India/South Asia, explains how Watson is being used in diverse industries like fashion as well. Designers at Falguni Shane Peacock took up IBM Watson to design their line called 'Future of Bollywood Fashion.' The system analysed data from 600,000 publicly available fashion runway images of the last decade, 5000 Bollywood celebrity images from social networks and 3000 historical Bollywood fashion images. With the help of visual recognition APIs and tools from the IBM Research Cognitive Fashion project, the system was able to assist significantly in the design process. **BY**

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